Chapter 28

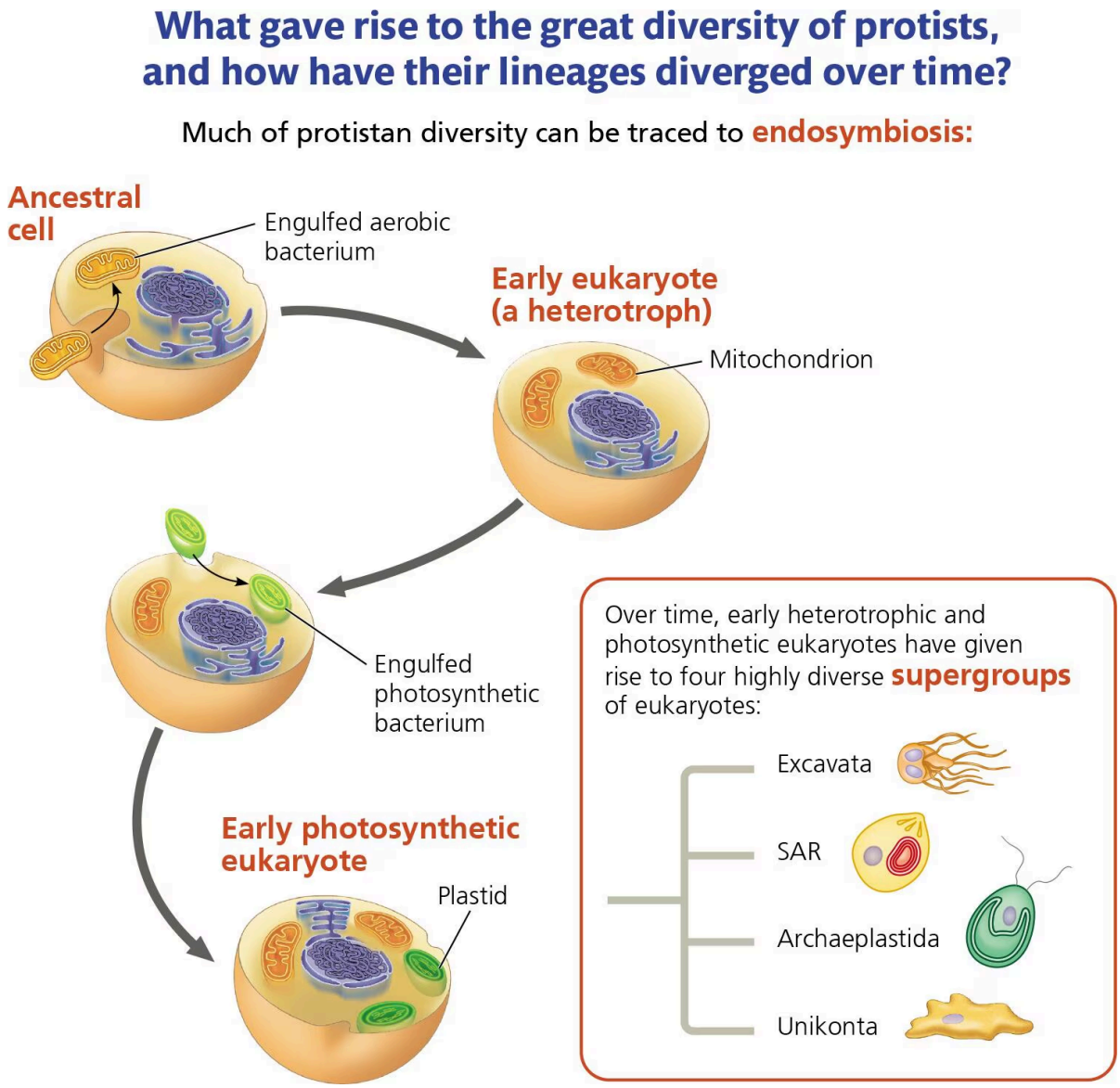
Campbell Biology, AP Edition, 12e

Chapter 28********

Protists

Figure 28.1

Knowing that prokaryotes are extremely small, you might assume that this image shows six prokaryotes and one much larger eukaryote. But in fact, the only prokaryote is the organism immediately above the scale bar—the rest are members of diverse, mostly unicellular groups of eukaryotes informally known as protists.



A schematic representation of the process of endosymbiosis attributing to protistan diversity is shown. The title is What gave rise to the great diversity of protists, and how have their lineages diverged over time? Much of protistan diversity can be traced to endosymbiosis. An ancestral cell has engulfed an aerobic bacterium and forms an early eukaryote (a heterotroph) with the engulfed aerobic bacterium labeled a mitochondrion. This eukaryotic cell engulfs a photosynthetic bacterium (in green) to form an early photosynthetic eukaryote. The engulfed photosynthetic bacterium is labeled a plastid.

A small box shows the highly diverse subgroups that eukaryotes are divided into: Excavata, SAR, Archaeplastida, and Unikonta, shown similar to a branching phylogenetic tree.

Key Concepts

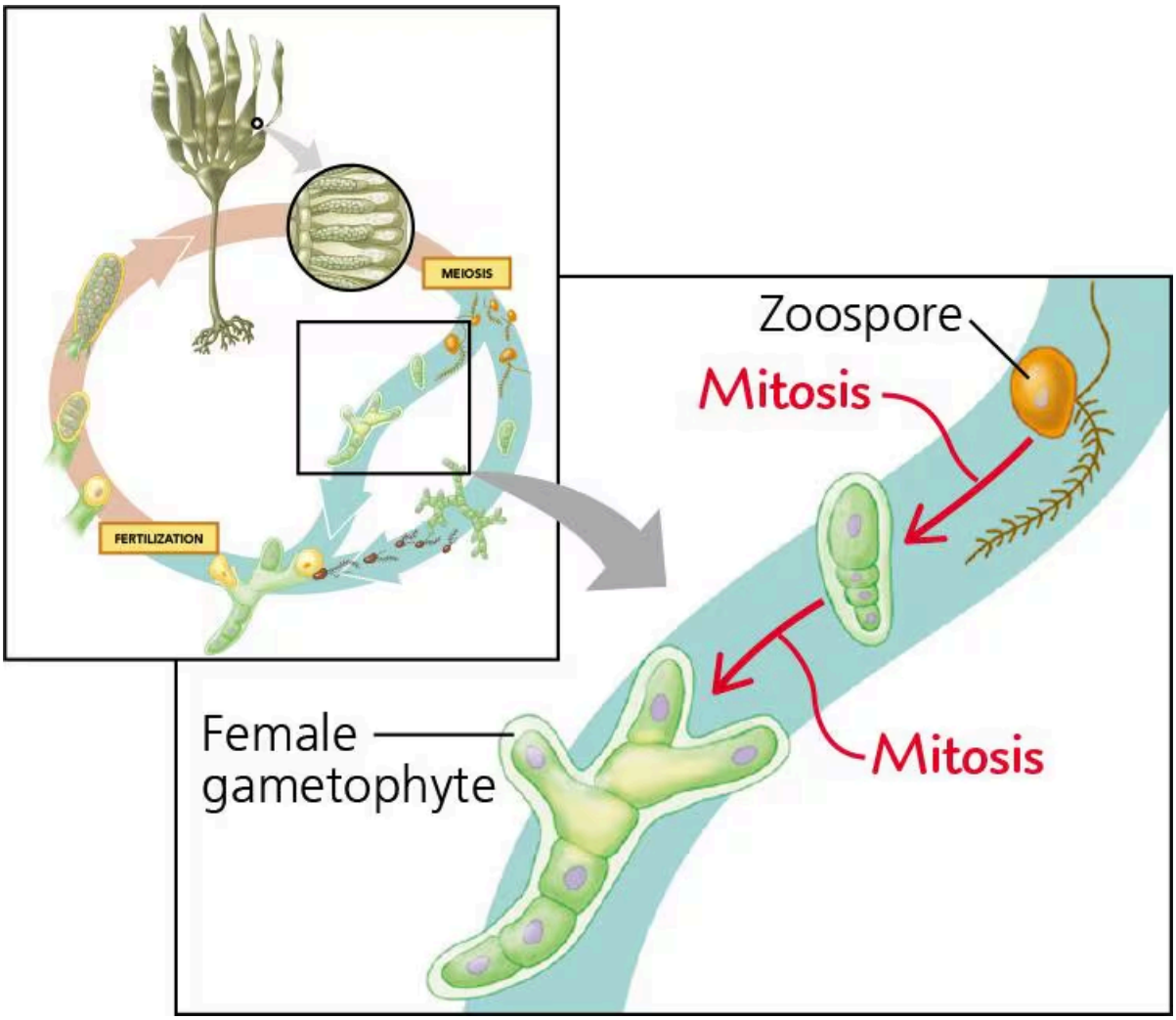
- 28.1**Most eukaryotes are single-celled organisms**

- 28.2**Excavates include protists with modified mitochondria and protists with unique flagella**
- 28.3**SAR is a highly diverse group of protists defined by DNA similarities**
- 28.4**Red algae and green algae are the closest relatives of plants**
- 28.5**Unikonts include protists that are closely related to fungi and animals**
- 28.6**Protists play key roles in ecological communities**

Protists represent a very diverse group of organisms. Structural evidence such as membrane-bound organelles **(BIG IDEAS 2 & 4)** and research techniques in biotechnology **(BIG IDEA 3)** support the relatedness of all eukaryotes **(BIG IDEA 1)**.

Study Tip

Label a drawing: Protist life cycles vary greatly. To make sense of this variation, it can be helpful to identify when mitosis occurs in different life cycles, as shown here for a portion of Figure 28.14. Identify the stages at which mitosis occurs in Figures 28.14, 28.18, 28.25, and 28.28.



The protist life cycle is depicted as a small diagram, with one section of it enlarged to show the steps from zoospores to the formation of female gametophytes. The mitotic steps in the cycle are to be labeled.

Get Ready for This Chapter

In this chapter, you will learn about the origin and evolutionary history of protists, as well as their impact on ecosystems and human societies. Before beginning this chapter, you should be able to do the following:

- Explain the concept of domains (see Concept 26.6, Figure 26.21).
- Describe the endosymbiotic origin of mitochondria and plastids (see Concept 25.3, Figure 25.10).
- Compare the outcomes of mitosis and meiosis (see Figure 13.10).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following domains consists of all the organisms whose cells have true nuclei?

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Animalia
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That's incorrect. Try again.

Incorrect. Review the following material: The domain Bacteria includes most of the currently known prokaryotes, while the domain Archaea consists of a diverse group of prokaryotic organisms that inhabit a wide variety of environments. The domain Eukarya includes many groups of single-celled organisms as well as plants, fungi, and animals.

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Bacteria
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That's incorrect. Try again.

Incorrect. Review the following material: The domain Bacteria includes most of the currently known prokaryotes, while the domain Archaea consists of a diverse group of prokaryotic organisms that inhabit a wide variety of environments. The domain Eukarya includes many groups of single-celled organisms as well as plants, fungi, and animals.

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Archaea
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That's incorrect. Try again.

Incorrect. Review the following material: The domain Bacteria includes most of the currently known prokaryotes, while the domain Archaea consists of a diverse group of prokaryotic organisms that inhabit a wide variety of environments. The domain Eukarya includes many groups of single-celled organisms as well as plants, fungi, and animals.

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Eukarya
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That's correct.

Correct! If you had trouble with this question, review the following material: The three domains—Bacteria, Archaea, and Eukarya—are a taxonomic level higher than the kingdom level. The domain Bacteria includes most of the currently known prokaryotes, while the domain Archaea consists of a diverse group of prokaryotic organisms that inhabit a wide variety of environments. The domain Eukarya consists of all the organisms that have cells containing true nuclei, including many groups of single-celled organisms as well as plants, fungi, and animals.

2. According to the hypothesis of serial endosymbiosis, which of the following was a key initial step that best describes the origin of chloroplasts?

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A prokaryotic host cell engulfed an aerobic, photosynthetic bacterium.
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That's incorrect. Try again.

Incorrect. Review the following material: Evidence indicates that the eukaryotes originated by endosymbiosis when a prokaryotic cell engulfed a small aerobic, heterotrophic bacterium whose descendants evolved into an organelle found in all eukaryotes, the mitochondrion. The prokaryotic ancestor of the mitochondrion probably entered the host cell as undigested prey or an internal parasite. Later, a eukaryotic host cell engulfed a photosynthetic bacterium, leading to the evolution of plastids.

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A eukaryotic host cell engulfed an aerobic, photosynthetic bacterium. (new correct
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mitochondrion probably entered the host cell as undigested prey or an internal parasite. Later, a eukaryotic host cell engulfed a photosynthetic bacterium, leading to the evolution of plastids.

3. A diploid (2n) cell divides by meiosis, producing four daughter cells. Next, each daughter cell divides by mitosis, producing a total of eight granddaughter cells. Which of the following statements about the granddaughter cells is true?

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All of the granddaughter cells are genetically identical.
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That's incorrect. Try again.

Incorrect. Review the following material: In meiosis, a parent cell produces four daughter cells, each of which has half the chromosome number of the parent cell. In mitosis, a parent cell produces two daughter cells, each of which is genetically identical to the parent cell.

Each granddaughter cell is diploid (2n).

That's incorrect. Try again.

Incorrect. Review the following material: In meiosis, a parent cell produces four daughter cells, each of which has half the chromosome number of the parent cell. In mitosis, a parent cell produces two daughter cells, each of which is genetically identical to the parent cell.

Each granddaughter cell is haploid (n).

That's correct.

Correct! If you had trouble with this question, review the following material: In meiosis, a parent cell produces four daughter cells, each of which has half the chromosome number of the parent cell; in this case, the diploid (2n) parent cell produces four daughter cells, each of which is haploid (n). In mitosis, a parent cell produces two daughter cells that are genetically identical to the parent cell; in this case, the four haploid daughter cells produce a total of eight granddaughter cells, each of which is haploid (n).

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Four of the granddaughter cells are genetically identical.
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That's incorrect. Try again.

Incorrect. Review the following material: In meiosis, a parent cell produces four daughter cells, each of which has half the chromosome number of the parent cell. In mitosis, a parent cell produces two daughter cells, each of which is genetically identical to the parent cell.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 28.1Describe the process of secondary endosymbiosis and explain its role in eukaryotic history.
- 28.2Characterize the excavates.
- 28.3Give examples of the protists classified in SAR.
- 28.4Describe characteristics of red and green algae.
- 28.5Identify and describe the closest eukaryotic relatives of fungi and animals.
- 28.6Give examples of ecological roles played by protists.